



IT-SIMPLERA INSTITUTE
Cyber Security Analyst Internship Program

WEEK 05 ASSIGNMENT

**Network Enumeration, Service Discovery, Vulnerability Assessment,
and Security Analysis**

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Table of Contents

Task 1 – Network Enumeration (15 Marks).....	4
Objective	4
Scan 1 – Default Nmap Scan	4
Command.....	4
Purpose	4
Results.....	4
Analysis	5
Scan 2 – Service Version Detection.....	5
Command.....	5
Purpose	5
Results.....	5
Analysis	6
Scan 3 – Operating System Detection.....	6
Command.....	6
Purpose	6
Results.....	6
Analysis	7
Scan 4 – Aggressive Scan	7
Command.....	7
Purpose	7
Results.....	7
Analysis	8
Scan 5 – Full Port Scan	8
Command.....	8
Purpose	8
Results.....	8
Analysis	8
Comparison of Nmap Scans.....	9
Task 2 – Service Enumeration	9
Objective:.....	9
Service Enumeration Table	10
Analysis:	11
Task 3 – Vulnerability Research.....	12
Objective	12
Vulnerability Research Table	12
Vulnerability Analysis.....	13
FTP Service (vsftpd 2.3.4).....	13

2. Apache HTTP Server 2.2.8.....	13
3. Samba 3.0.20	13
Summary	13
Task 4 – Scan Comparison.....	14
Objective	14
Comparison of Nmap Scans	14
Question & Answer	14
Task 5 – Security Analysis & Remediation Recommendations.....	15
Objective	15
Security Analysis & Remediation Table.....	15
Detailed Security Assessment.....	17
1. FTP Service (Port 21)	17
2. SSH Service (Port 22).....	18
3. Telnet Service (Port 23)	19
4. Apache HTTP Server (Port 80).....	19
5. Samba Service (Port 445)	20
6. MySQL Service (Port 3306)	21
7. Apache Tomcat (Port 8180).....	21
Task 6 – Vulnerability Assessment Report	22
Introduction	22
Scope.....	22
Tools Used	22
Enumeration Results.....	23
Scans Performed	23
Summary of Findings	23
Vulnerability Findings.....	23
Screenshots.....	24
Default Nmap Scan	24
Service Version Scan (-sV)	25
Operating System Detection (-OS)	25
Aggressive Scan (-A).....	26
Full Port Scan (-p-)	26
Recommendations	27
Challenges Faced.....	27
Conclusion.....	27
References	27

Task 1 – Network Enumeration (15 Marks)

Objective

The objective of this task was to perform network enumeration on the assigned target machine using different Nmap scanning techniques. The purpose of network enumeration is to identify active hosts, discover open ports, detect running services, determine service versions, identify the operating system, and gather information that may help in assessing the target's security posture. Five different Nmap scan types were performed to compare their capabilities and effectiveness.

Target IP Address: 192.168.0.107

Scanning Machine: Kali Linux

Tool Used: Nmap 7.95

Scan 1 – Default Nmap Scan

Command

```
nmap 192.168.0.107
```

Purpose

The default Nmap scan examines the 1000 most common TCP ports on the target system to identify open ports and the services associated with them. This scan is fast and provides a quick overview of the target's exposed services.

Results

- **Host Status:** Up
- **Scan Time:** 0.42 seconds
- **Closed Ports:** 977
- **Open Ports:** 23

Port	Protocol	Service
21	TCP	FTP
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP	DNS
80	TCP	HTTP
111	TCP	RPCBind
139	TCP	NetBIOS-SSN
445	TCP	Microsoft-DS
512	TCP	exec
513	TCP	login
514	TCP	shell
1099	TCP	Java RMI
1524	TCP	Bind Shell
2049	TCP	NFS
2121	TCP	FTP
3306	TCP	MySQL
5432	TCP	PostgreSQL
5900	TCP	VNC
6000	TCP	X11
6667	TCP	IRC
8009	TCP	AJP13
8180	TCP	HTTP

Analysis

The default scan quickly identified 23 open TCP ports running various network services. However, it did not provide service versions or operating system information. This scan is useful for obtaining a quick overview of the target's attack surface.

```
(kali@kali)-[~]
$ nmap 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:45 EDT
Nmap scan report for 192.168.0.107
Host is up (0.0024s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:18:B3:56 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds
```

Scan 2 – Service Version Detection

Command

```
nmap -sV 192.168.0.107
```

Purpose

The **-sV** option enables service version detection. It identifies the software and version running on each open port, which is essential for vulnerability assessment because known vulnerabilities are usually associated with specific software versions.

Results

Port	Service	Version
21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH 4.7p1 Debian 8ubuntu1
23	Telnet	Linux telnetd
25	SMTP	Postfix smtpd
53	DNS	ISC BIND 9.4.2
80	HTTP	Apache HTTPD 2.2.8
111	RPCBind	Version 2
139	NetBIOS	Samba 3.x-4.x
445	SMB	Samba 3.0.20
512	exec	netkit-rsh rexecd
513	login	rlogind

1099	Java RMI	GNU Classpath grmiregistry
1524	Bind Shell	Metasploitable Root Shell
2049	NFS	Version 2–4
2121	FTP	ProFTPD 1.3.1
3306	MySQL	5.0.51a
5432	PostgreSQL	8.3.x
5900	VNC	Protocol 3.3
6667	IRC	UnrealIRCd
8009	AJP13	Apache JServ
8180	HTTP	Apache Tomcat 5.5

Analysis

The service version scan identified the exact versions of the software running on the target machine. Several outdated services such as **vsftpd 2.3.4**, **Apache HTTP Server 2.2.8**, **Samba 3.0.20**, and **UnrealIRCd** were detected. These versions are known to contain publicly documented vulnerabilities, making them important targets for further security analysis.

```
(kali@kali)-[~]
└─$ nmap -sV 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:45 EDT
Nmap scan report for 192.168.0.107
Host is up (0.0032s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache JServ (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:18:B3:56 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.00 seconds
(kali@kali)-[~]
```

Scan 3 – Operating System Detection

Command

```
sudo nmap -O 192.168.0.107
```

Purpose

The **-O** option attempts to identify the target operating system by analyzing TCP/IP stack characteristics.

Results

- Operating System: Linux
- Kernel Version: Linux 2.6.9 – 2.6.33
- Device Type: General Purpose
- Network Distance: 1 Hop

Analysis

The scan successfully identified the target as a Linux system running an older Linux 2.6 kernel. Older operating systems often contain security vulnerabilities that have been addressed in newer releases. Operating system detection helps security professionals understand the environment and select appropriate testing techniques.

```
(kali㉿kali)-[~]
└─$ sudo nmap -O 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:46 EDT
Nmap scan report for 192.168.0.107
Host is up (0.0021s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:18:B3:56 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.12 seconds
```

Scan 4 – Aggressive Scan

Command

```
sudo nmap -A 192.168.0.107
```

Purpose

The **-A** option performs an aggressive scan that combines operating system detection, service version detection, default NSE script execution, and traceroute.

Results

The aggressive scan provided additional information, including:

- Anonymous FTP login enabled
- FTP server information
- SSH host keys
- SMTP supported commands
- DNS version disclosure
- Apache web server title
- SMB security configuration
- MySQL server information
- PostgreSQL SSL certificate details
- IRC server information
- Apache Tomcat details
- NetBIOS information

- Traceroute results
- Operating system fingerprint

Analysis

The aggressive scan produced the most comprehensive results among all scans. It revealed several potential security weaknesses, including anonymous FTP access, outdated software versions, SMB message signing disabled, and detailed service information. These findings provide valuable information for vulnerability assessment and penetration testing.

```
(kali@kali)~$ sudo nmap -A 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:47 EDT
Nmap scan report for 192.168.0.107
Host is up (0.00077s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to 192.168.0.106
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_  vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp_commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_  SSL2_RC2_128_CBC_WITH_MD5
|_  SSL2_DES_64_CBC_WITH_MD5
|_  SSL2_RC4_128_EXPORT40_WITH_MD5
|_  SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_  SSL2_RC4_128_WITH_MD5
|_  SSL2_DES_192_EDE3_CBC_WITH_MD5
|_ssl-date: 2026-08-01T01:47:37+00:00; 0s from scanner time.
53/tcp    open  domain       ISC BIND 9.4.2
```

Scan 5 – Full Port Scan

Command

`nmap -p- 192.168.0.107`

Purpose

The **-p-** option scans all 65,535 TCP ports instead of only the default 1000 ports. This helps identify services running on uncommon or high-numbered ports.

Results

The scan discovered **29 open TCP ports**, including additional high-numbered ports that were not identified during the default scan.

Additional ports discovered include:

Port	Service
3632	distccd
6697	IRCS-U
32838	unknown
41224	unknown
57716	unknown
59591	unknown

Analysis

The full port scan discovered several additional services running on non-standard ports. These services increase the attack surface and may expose additional vulnerabilities. Performing a full port

scan is recommended during comprehensive security assessments because important services may be running outside the default top 1000 ports.

```
(kali@kali)-[~]
$ nmap -p- 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:47 EDT
Nmap scan report for 192.168.0.107
Host is up (0.0033s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
32838/tcp open  unknown
41224/tcp open  unknown
57716/tcp open  unknown
59591/tcp open  unknown
MAC Address: 00:0C:29:18:B3:56 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 7.49 seconds
```

Comparison of Nmap Scans

Scan Command	Purpose	Scan Time	Information Collected
nmap	Default scan	0.42 sec	Open ports
nmap -sV	Service version detection	12 sec	Open ports and software versions
nmap -O	Operating system detection	2.12 sec	Open ports and operating system
nmap -A	Aggressive scan	26.96 sec	OS, versions, NSE scripts, traceroute, additional security information
nmap -p-	Full TCP port scan	7.49 sec	All 65,535 TCP ports

Task 2 – Service Enumeration

Objective:

The objective of this task was to enumerate all services running on the target machine using the results obtained from the **Nmap Service Version Detection (nmap -sV)** scan. Service enumeration helps identify the software, versions, and purpose of each service running on open ports. This information is essential for vulnerability assessment because outdated or misconfigured services may expose the system to security risks.

Service Enumeration Table

Port Number	Protocol	Service Name	Service Version	Port Status	Purpose of the Service
21	TCP	FTP	vsftpd 2.3.4	Open	Allows file transfer between client and server.
22	TCP	SSH	OpenSSH 4.7p1 Debian 8ubuntu1	Open	Provides secure remote login and command execution.
23	TCP	Telnet	Linux telnetd	Open	Provides remote command-line access without encryption.
25	TCP	SMTP	Postfix smtpd	Open	Handles sending and receiving email messages.
53	TCP	DNS	ISC BIND 9.4.2	Open	Resolves domain names into IP addresses.
80	TCP	HTTP	Apache HTTP Server 2.2.8 (Ubuntu) DAV/2	Open	Hosts websites and web applications.
111	TCP	RPCBind	Version 2	Open	Maps RPC services to network ports.
139	TCP	NetBIOS-SSN	Samba smbd 3.x-4.x	Open	Supports file and printer sharing using NetBIOS.
445	TCP	Microsoft-DS (SMB)	Samba smbd 3.0.20-Debian	Open	Provides SMB file and printer sharing services.
512	TCP	exec	netkit-rsh rexecd	Open	Allows remote execution of commands using the RSH protocol.
513	TCP	login	OpenBSD/Solaris rlogind	Open	Enables remote login using the Rlogin protocol.
514	TCP	shell	tcpwrapped	Open	Provides remote shell services and access control.
1099	TCP	Java RMI	GNU Classpath grmiregistry	Open	Enables Java Remote Method Invocation communication.
1524	TCP	Bind Shell	Metasploitable Root Shell	Open	Provides a root shell listening on a TCP port (intentional vulnerability).
2049	TCP	NFS	NFS Version 2-4	Open	Allows remote file system sharing across Linux systems.

2121	TCP	FTP	ProFTPD 1.3.1	Open	Provides an alternative FTP service for file transfers.
3306	TCP	MySQL	MySQL 5.0.51a-3ubuntu5	Open	Database server used by web applications.
5432	TCP	PostgreSQL	PostgreSQL 8.3.0–8.3.7	Open	Open-source relational database management system.
5900	TCP	VNC	VNC Protocol 3.3	Open	Provides graphical remote desktop access.
6000	TCP	X11	Access Denied	Open	Supports remote graphical display for Linux systems.
6667	TCP	IRC	UnrealIRCd	Open	Internet Relay Chat server for real-time communication.
8009	TCP	AJP13	Apache JServ Protocol v1.3	Open	Enables communication between Apache HTTP Server and Apache Tomcat.
8180	TCP	HTTP	Apache Tomcat/Coyote JSP Engine 1.1	Open	Hosts Java web applications and JSP pages.

Analysis:

The service enumeration identified **23 open TCP ports** running a variety of network services, including file transfer, remote administration, web hosting, email, databases, and network file sharing. Several services are running outdated software versions, including **vsftpd 2.3.4**, **Apache HTTP Server 2.2.8**, **Samba 3.0.20**, **MySQL 5.0.51a**, **PostgreSQL 8.3**, and **UnrealIRCd**. Additionally, insecure services such as **Telnet**, **RSH**, **Rlogin**, and an intentionally exposed **bind shell (port 1524)** were identified. These services significantly increase the target system's attack surface and provide multiple opportunities for exploitation. The information gathered during this phase will be used in the next task to research known vulnerabilities (CVEs), assess security risks, and recommend appropriate remediation measures.

```
(kali@kali)-[~]
$ nmap -sV 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:45 EDT
Nmap scan report for 192.168.0.107
Host is up (0.0032s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:18:B3:56 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.00 seconds
(kali@kali)-[~]
```

Task 3 – Vulnerability Research

Objective

The objective of this task was to research publicly disclosed vulnerabilities associated with the services identified during the service enumeration phase. The discovered software versions were compared with the National Vulnerability Database (NVD), MITRE CVE database, and vendor security advisories to identify known Common Vulnerabilities and Exposures (CVEs). This process helps security analysts understand the risks associated with outdated software and recommend appropriate remediation measures.

Vulnerability Research Table

Service	Version	CVE ID	Vulnerability Name	CVSS Score	Severity	Short Description	Recommended Fix
FTP (vsftpd)	2.3.4	CVE-2011-2523	vsftpd Backdoor Command Execution	10.0	Critical	A malicious backdoor was introduced into the vsftpd 2.3.4 release, allowing attackers to gain remote shell access without authentication.	Upgrade to the latest secure version of vsftpd and disable the vulnerable release.
Apache HTTP Server	2.2.8	CVE-2011-3192	Apache Range Header Denial of Service	7.8	High	A specially crafted HTTP Range request can consume excessive server resources, causing a	Upgrade Apache to a supported version and apply the latest security patches.

						denial-of-service (DoS).	
Samba (SMB)	3.0.20	CVE-2007-2447	Samba "username map script" Command Execution	10.0	Critical	Improper handling of the "username map script" option allows authenticated attackers to execute arbitrary commands with root privileges.	Upgrade Samba to a supported version and disable the vulnerable configuration.

Vulnerability Analysis

FTP Service (vsftpd 2.3.4)

The FTP service is running **vsftpd version 2.3.4**, which is affected by **CVE-2011-2523**. This version was compromised by a malicious backdoor that allows attackers to obtain remote shell access by sending a specially crafted username. Since this vulnerability enables remote code execution without authentication, it poses a **Critical** security risk. Upgrading to a newer version of vsftpd and removing the vulnerable release completely mitigates this issue.

2. Apache HTTP Server 2.2.8

The web server is running **Apache HTTP Server 2.2.8**, which is vulnerable to **CVE-2011-3192**. An attacker can exploit this flaw by sending specially crafted HTTP Range headers, causing excessive memory consumption and potentially making the web server unavailable. This vulnerability is rated **High** due to its impact on service availability. Updating Apache to the latest supported version and applying vendor security patches effectively resolves this issue.

3. Samba 3.0.20

The SMB service is running **Samba version 3.0.20**, which is vulnerable to **CVE-2007-2447**. This vulnerability allows attackers to execute arbitrary commands with root privileges through the "username map script" feature if it is enabled. Successful exploitation results in complete system compromise, making it a **Critical** vulnerability. The recommended mitigation is to upgrade Samba to a supported version and disable the vulnerable configuration option.

Summary

The vulnerability assessment identified several outdated services running on the target machine. Among the researched services, **vsftpd 2.3.4** and **Samba 3.0.20** present **Critical** security risks because they can lead to remote code execution and complete system compromise. **Apache HTTP Server 2.2.8** is affected by a **High** severity denial-of-service vulnerability that can disrupt service availability. These findings demonstrate the importance of keeping software updated, disabling unnecessary services, and regularly applying security patches to reduce the system's attack surface.

Task 4 – Scan Comparison

Objective

The objective of this task was to compare the results obtained from different Nmap scanning techniques. Each scan provides different levels of information depending on its purpose. By comparing these scans, it is possible to determine which scan is most suitable for various stages of a security assessment and understand the advantages and limitations of each scanning method.

Comparison of Nmap Scans

Scan Command	Purpose	Scan Time	Information Collected	Advantages	Disadvantages
nmap 192.168.0.107	Default Scan	0.42 sec	Identified the most common open TCP ports.	Very fast and suitable for initial reconnaissance.	Does not identify service versions or operating system.
nmap -sV 192.168.0.107	Service Version Detection	12.00 sec	Detected service names and software versions running on each open port.	Helps identify outdated software and known vulnerabilities.	Slower than the default scan.
sudo nmap -O 192.168.0.107	Operating System Detection	2.12 sec	Identified the target operating system as Linux Kernel 2.6.x.	Useful for understanding the target environment.	OS detection may not always be accurate and requires root privileges.
sudo nmap -A 192.168.0.107	Aggressive Scan	26.96 sec	Collected service versions, operating system details, NSE script results, traceroute, host information, and security configurations.	Provides the most comprehensive information in a single scan.	Slowest scan and more likely to be detected by intrusion detection systems.
nmap -p- 192.168.0.107	Full Port Scan	7.49 sec	Scanned all 65,535 TCP ports and discovered additional high-numbered open ports.	Detects services running on uncommon ports that may be missed by default scans.	Does not identify service versions unless combined with -sV.

Question & Answer

Which scan was the fastest?

nmap 192.168.0.107

It completed in approximately **0.42 seconds**, making it the fastest scan.

Which scan discovered the most information?

sudo nmap -A 192.168.0.107

It identified:

- Service versions
- Operating system
- NSE script results
- SMB information
- FTP anonymous access
- DNS details
- HTTP titles
- Traceroute

Which scan would you use in a real-world assessment and why?

I would use **nmap -A** because it combines service version detection, operating system detection, NSE script execution, and traceroute into a single scan. It provides detailed information necessary for identifying vulnerabilities and planning further security assessments.

Advantages and Disadvantages of Each Scan

Scan	Advantages	Disadvantages
nmap	Fast and lightweight	Limited information
nmap -sV	Detects service versions for CVE research	Takes longer
nmap -O	Identifies target operating system	Requires elevated privileges and may be inaccurate
nmap -A	Provides the most detailed information	Slow and easily detected by security monitoring
nmap -p-	Finds services on non-standard ports	Longer scan and lacks version detection without -sV

Task 5 – Security Analysis & Remediation Recommendations

Objective

The objective of this task was to perform a detailed security assessment of the services discovered during the network enumeration phase. Each identified service was analyzed to determine its purpose, potential security risks, possible attack scenarios, risk level, remediation techniques, verification methods, and security best practices. The recommendations are based on industry standards such as **NIST**, **OWASP**, **CIS Benchmarks**, and official vendor documentation.

Security Analysis & Remediation Table

Open Port / Service	Security Risk	Attack Scenario	Risk Level	Remediation	Verification Method	Reference
21 – FTP (vsftpd 2.3.4)	Outdated FTP server with anonymous login and known backdoor vulnerability (CVE-2011-2523).	An attacker logs in anonymously or exploits the backdoor to gain unauthorized shell access and full control	Critical	1. Upgrade to the latest version of vsftpd. 2. Disable anonymous FTP access. 3. Restrict FTP access using firewall rules and strong	Perform an Nmap version scan (nmap -sV) to verify the updated version and confirm anonymou	NIST, vsftpd Security Advisory

		of the server.		authentication.	s login is disabled.	
22 – SSH (OpenSSH 4.7p1)	Running an outdated SSH version that may contain known security vulnerabilities.	An attacker performs brute-force attacks or exploits weaknesses in the outdated SSH service to gain remote access.	High	1. Upgrade OpenSSH to the latest supported version. 2. Disable root login. 3. Use SSH key authentication and Multi-Factor Authentication (MFA).	Verify the new OpenSSH version using nmap -sV and confirm root login is disabled.	CIS SSH Benchmark, OpenSSH Documentation
23 – Telnet	Telnet transmits usernames and passwords in plain text without encryption.	An attacker captures network traffic using packet sniffing tools to steal user credentials.	High	1. Disable the Telnet service. 2. Replace Telnet with SSH. 3. Restrict remote administration using firewall rules.	Verify that port 23 is closed using Nmap and ensure SSH is available instead.	NIST SP 800-123
80 – Apache HTTP Server 2.2.8	Outdated Apache version vulnerable to known attacks such as CVE-2011-3192.	An attacker sends crafted HTTP requests to cause denial of service or exploit web server vulnerabilities.	High	1. Upgrade Apache to the latest supported version. 2. Apply vendor security patches. 3. Enable HTTPS and deploy a Web Application Firewall (WAF).	Perform an Nmap version scan and verify the updated Apache version.	Apache Security Advisories, OWASP
445 – Samba 3.0.20	Vulnerable SMB service that may allow remote code execution (CVE-2007-2447).	An attacker exploits the vulnerable Samba service to execute arbitrary commands and gain root privileges.	Critical	1. Upgrade Samba to the latest supported version. 2. Disable unnecessary SMB shares. 3. Enable SMB signing and restrict network access.	Confirm the Samba version has been updated using nmap -sV and verify SMB signing is enabled.	Samba Security Advisories

3306 – MySQL 5.0.51a	Outdated database server with potential authentication and privilege escalation vulnerabilities.	An attacker gains unauthorized access using weak credentials and steals or modifies sensitive data.	Medium	1. Upgrade MySQL to a supported version. 2. Use strong passwords and least-privilege accounts. 3. Restrict database access to trusted hosts only.	Verify the MySQL version and ensure remote access is restricted.	MySQL Security Documentation
8180 – Apache Tomcat 5.5	Outdated application server with multiple publicly known vulnerabilities.	An attacker exploits a vulnerable web application or default configuration to execute malicious code.	High	1. Upgrade Apache Tomcat to the latest supported version. 2. Remove default applications and accounts. 3. Configure secure authentication and HTTPS.	Verify the updated Tomcat version and confirm default applications are removed.	Apache Tomcat Security Advisories

Detailed Security Assessment

1. FTP Service (Port 21)

Purpose

The FTP service is used to transfer files between systems over a network.

Security Risk

The server is running **vsftpd 2.3.4**, which is affected by a critical backdoor vulnerability. Anonymous FTP login is also enabled, allowing unauthorized users to access the server.

Attack Scenario

An attacker can exploit the backdoor vulnerability or log in anonymously to gain unauthorized access to the server and potentially obtain a remote shell.

Risk Level

Critical

Remediation

- Upgrade vsftpd to the latest secure version.

- Disable anonymous FTP login.
- Restrict FTP access using firewall rules and strong authentication.

Verification

Run:

`nmap -sV 192.168.0.107`

Confirm that the server is running a patched version and anonymous login is disabled.

Reference

- NIST National Vulnerability Database
- vsftpd Security Advisory

2. SSH Service (Port 22)

Purpose

SSH provides secure remote administration of Linux systems.

Security Risk

The target is running an outdated version of OpenSSH that may contain known vulnerabilities.

Attack Scenario

An attacker may perform brute-force attacks or exploit weaknesses in the outdated SSH implementation to gain unauthorized access.

Risk Level

High

Remediation

- Upgrade OpenSSH.
- Disable root login.
- Use SSH key authentication and MFA.

Verification

Verify the updated version using:

`nmap -sV 192.168.0.107`

Attempt a root login to ensure it is disabled.

Reference

- CIS Benchmarks
- OpenSSH Documentation

3. Telnet Service (Port 23)

Purpose

Telnet provides remote command-line access.

Security Risk

Telnet transmits all communication, including usernames and passwords, in plain text.

Attack Scenario

An attacker connected to the same network can capture Telnet traffic using packet sniffing tools and steal login credentials.

Risk Level

High

Remediation

- Disable the Telnet service.
- Replace it with SSH.
- Restrict remote access using firewall rules.

Verification

Run:

```
nmap 192.168.0.107
```

Verify that port **23** is no longer open.

Reference

- NIST SP 800-123

4. Apache HTTP Server (Port 80)

Purpose

Apache hosts websites and web applications.

Security Risk

The server is running Apache 2.2.8, which is outdated and vulnerable to multiple publicly disclosed vulnerabilities.

Attack Scenario

An attacker can exploit known Apache vulnerabilities to cause denial-of-service attacks or compromise web applications.

Risk Level

High

Remediation

- Upgrade Apache to the latest supported version.
- Apply all security patches.
- Enable HTTPS and deploy a Web Application Firewall (WAF).

Verification

Use:

```
nmap -sV 192.168.0.107
```

Verify the updated Apache version.

Reference

- Apache Security Advisories
- OWASP Web Security Guidelines

5. Samba Service (Port 445)

Purpose

Samba provides file and printer sharing between Linux and Windows systems.

Security Risk

Samba 3.0.20 is vulnerable to remote code execution.

Attack Scenario

An attacker exploits the vulnerable Samba service to execute arbitrary commands and obtain root privileges.

Risk Level

Critical

Remediation

- Upgrade Samba.
- Disable unnecessary shares.
- Enable SMB signing and restrict access.

Verification

Perform another version scan and confirm SMB signing is enabled.

Reference

- Samba Security Advisories

6. MySQL Service (Port 3306)

Purpose

MySQL stores and manages application databases.

Security Risk

The outdated MySQL version may contain authentication and privilege escalation vulnerabilities.

Attack Scenario

An attacker obtains database credentials through brute-force attacks or weak passwords and steals sensitive information.

Risk Level

Medium

Remediation

- Upgrade MySQL.
- Enforce strong passwords.
- Limit database access to trusted hosts.

Verification

Confirm the updated version using Nmap and verify that remote database access is restricted.

Reference

- MySQL Security Documentation

7. Apache Tomcat (Port 8180)

Purpose

Apache Tomcat hosts Java-based web applications.

Security Risk

The server is running an outdated Tomcat version that may contain known vulnerabilities.

Attack Scenario

An attacker exploits vulnerable web applications or default configurations to execute malicious code on the server.

Risk Level

High

Remediation

- Upgrade Apache Tomcat.
- Remove default applications and accounts.
- Enable HTTPS and strong authentication.

Verification

Verify the Tomcat version using `nmap -sV` and ensure default applications have been removed.

Reference

- Apache Tomcat Security Advisories

Task 6 – Vulnerability Assessment Report

Introduction

Network enumeration and vulnerability assessment are essential phases of the cybersecurity assessment process. Enumeration helps identify active hosts, open ports, running services, software versions, and operating systems, while vulnerability assessment identifies weaknesses that could be exploited by attackers. In this assignment, Nmap was used to enumerate the services running on the target machine (Metasploitable 2). The discovered services were analyzed to identify known vulnerabilities (CVEs), evaluate their security risks, and recommend appropriate remediation measures based on industry best practices.

Scope

The scope of this assessment was limited to the target machine with IP address **192.168.0.107**. The assessment included:

- Network enumeration
- Open port identification
- Service enumeration
- Operating system detection
- Service version detection
- Vulnerability research
- Security risk analysis
- Remediation recommendations

No exploitation of vulnerabilities was performed during this assessment.

Tools Used

Tool	Purpose
------	---------

Kali Linux	Penetration Testing Operating System
Nmap 7.95	Network Enumeration and Port Scanning
Internet Browser	CVE and Vulnerability Research
Microsoft Word	Documentation
Draw.io	Network Diagram
PDF Converter	Report Generation

Enumeration Results

Five different Nmap scans were performed on the target machine to gather information about open ports, running services, software versions, and the operating system.

Scans Performed

- nmap 192.168.0.107
- nmap -sV 192.168.0.107
- sudo nmap -O 192.168.0.107
- sudo nmap -A 192.168.0.107
- nmap -p- 192.168.0.107

Summary of Findings

- Host Status: Up
- Operating System: Linux Kernel 2.6.x
- Open Ports Discovered: 29
- Major Services:
 - FTP
 - SSH
 - Telnet
 - SMTP
 - DNS
 - HTTP
 - SMB
 - NFS
 - MySQL
 - PostgreSQL
 - VNC
 - IRC
 - Apache Tomcat

The Aggressive Scan (nmap -A) provided the most detailed information, including service versions, operating system details, NSE script results, and security configurations.

Vulnerability Findings

The assessment identified multiple outdated and vulnerable services running on the target machine.

Service	Version	Vulnerability	Risk Level
vsftpd	2.3.4	CVE-2011-2523	Critical
Samba	3.0.20	CVE-2007-2447	Critical
Apache HTTP Server	2.2.8	CVE-2011-3192	High
OpenSSH	4.7p1	Outdated Version	High
Telnet	Linux telnetd	Plain Text Communication	High

MySQL	5.0.51a	Outdated Version	Medium
Apache Tomcat	5.5	Outdated Version	High

These vulnerabilities increase the attack surface and could allow attackers to gain unauthorized access, execute arbitrary commands, steal sensitive data, or disrupt system availability.

Screenshots

Include the following screenshots in this section:

Default Nmap Scan

```
(kali@kali)-[~]
$ nmap 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:45 EDT
Nmap scan report for 192.168.0.107
Host is up (0.0024s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:18:B3:56 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds
```

Service Version Scan (-sV)

```
(kali@kali)-[~]
$ nmap -sV 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:45 EDT
Nmap scan report for 192.168.0.107
Host is up (0.0032s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:18:B3:56 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.00 seconds

(kali@kali)-[~]
```

Operating System Detection (-o)S

```
(kali@kali)-[~]
$ sudo nmap -o 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:46 EDT
Nmap scan report for 192.168.0.107
Host is up (0.0021s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:18:B3:56 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.12 seconds
```

Aggressive Scan (-A)

```
(kali@kali)-[~]
$ sudo nmap -A 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:47 EDT
Nmap scan report for 192.168.0.107
Host is up (0.00077s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_   Connected to 192.168.0.106
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_sslv2:
|_sslv2 supported
|_ciphers:
|_   SSL2_RC2_128_CBC_WITH_MD5
|_   SSL2_DES_64_CBC_WITH_MD5
|_   SSL2_RC4_128_EXPORT40_WITH_MD5
|_   SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_   SSL2_RC4_128_WITH_MD5
|_   SSL2_DES_192_EDE3_CBC_WITH_MD5
|_ssl-date: 2026-08-01T01:47:37+00:00; 0s from scanner time.
53/tcp    open  domain       ISC BIND 9.4.2
```

Full Port Scan (-p-)

```
(kali@kali)-[~]
$ nmap -p- 192.168.0.107
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 21:47 EDT
Nmap scan report for 192.168.0.107
Host is up (0.0033s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
32838/tcp open  unknown
41224/tcp open  unknown
57716/tcp open  unknown
59591/tcp open  unknown
MAC Address: 00:0C:29:18:B3:56 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 7.49 seconds
```


Recommendations

The following security recommendations are proposed to improve the security posture of the target system:

- Upgrade all outdated software to the latest supported versions.
- Disable anonymous FTP access.
- Remove insecure services such as Telnet and replace them with SSH.
- Disable unnecessary services and close unused ports.
- Enable SMB message signing.
- Configure firewall rules to restrict access to critical services.
- Use strong passwords and implement Multi-Factor Authentication (MFA).
- Enable HTTPS instead of HTTP for web applications.
- Regularly perform vulnerability scans and penetration testing.
- Apply security patches promptly and follow vendor security advisories.
- Follow CIS Benchmarks, OWASP recommendations, and NIST security guidelines for system hardening.

Challenges Faced

During the assessment, several challenges were encountered:

- Identifying outdated software versions and mapping them to the correct CVEs.
- Interpreting the detailed output generated by the Aggressive Scan.
- Differentiating between standard and non-standard services.
- Researching vulnerabilities and selecting the most relevant security advisories.
- Understanding the purpose of each service and evaluating its associated security risks.

These challenges were addressed through careful analysis of the Nmap output and consultation of trusted security resources.

Conclusion

The vulnerability assessment successfully identified numerous exposed services and outdated software running on the target machine. The scans revealed **29 open TCP ports**, several legacy protocols, and multiple services with publicly known vulnerabilities. Critical findings included **vsftpd 2.3.4**, **Samba 3.0.20**, and **Apache HTTP Server 2.2.8**, all of which present significant security risks if left unpatched. The assessment highlights the importance of regular network scanning, timely software updates, secure system configuration, and continuous vulnerability management. Implementing the recommended remediation measures will significantly reduce the attack surface and strengthen the overall security posture of the system.

References

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